Regression

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Load Packages

```
library(MASS)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:MASS':
##
##     select
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(psych)
```

```
##
## Attaching package: 'psych'
```

```
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```
library(ppcor)
```

Disable Scientific Notation

```
options(scipen = 999)
```

A researcher is interested in examining whether reading skills (READING) at kindergarten entry is predicted by cognitive stimulation (COGSTIM) in the home environment and is concerned about the confounding influence of children's age when they start kindergarten (AGEENTRY).

Disable Scientific Notation

```
options(scipen = 999)
```

Read in data

```
data_1 <- read.csv("regression_df.csv")
```

```
head(data_1)
```

```
##    CHILDID AGEENTRY BLACK COGSTIM READING
## 1 0001002C       74     0    2.68   45.17
## 2 0001007C       60     0    2.34   25.67
## 3 0001010C       67     0    2.91   27.34
## 4 0002003C       71     0    2.25   42.51
## 5 0002004C       63     0    2.61   36.15
## 6 0002006C       59     0    2.31   22.59
```

Variables: - READING = reading skills (DV) - COGSTIM = cognitive stimulation (IV) - AGEENTRY = are when children start kindergarten (IV)

Let's calculate the regression equation predicting reading skills with cognitive stimulation and age at kindergarten entry. Present the standardized and unstandardized regression equations.

Unstandardized

```
model_1 <-  lm(READING ~ COGSTIM + AGEENTRY, data= data_1)
summary(model_1)
```

```
##
## Call:
## lm(formula = READING ~ COGSTIM + AGEENTRY, data = data_1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.6189  -4.7406  -0.9091   3.9586  26.1751
##
## Coefficients:
##             Estimate Std. Error t value             Pr(>|t|)
## (Intercept) -3.59421    1.36613  -2.631              0.00854 **
## COGSTIM      6.72643    0.19163  35.101 < 0.0000000000000002 ***
## AGEENTRY     0.29179    0.01971  14.808 < 0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.649 on 6396 degrees of freedom
## Multiple R-squared:  0.1839, Adjusted R-squared:  0.1837
## F-statistic: 720.7 on 2 and 6396 DF,  p-value: < 0.00000000000000022
```

```
anova(model_1)
```

```
## Analysis of Variance Table
##
## Response: READING
##             Df Sum Sq Mean Sq F value                Pr(>F)
## COGSTIM      1  54027   54027 1222.22 < 0.00000000000000022 ***
## AGEENTRY     1   9693    9693  219.27 < 0.00000000000000022 ***
## Residuals 6396 282726      44
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
confint(model_1,level=0.95)
```

```
##                  2.5 %     97.5 %
## (Intercept) -6.2722848 -0.9161269
## COGSTIM      6.3507773  7.1020879
## AGEENTRY     0.2531597  0.3304168
```

```
coefficients(model_1)
```

```
## (Intercept)     COGSTIM    AGEENTRY
##  -3.5942059   6.7264326   0.2917882
```

The intercept is -3.59 and is the predicted level of reading skills (READING) when cognitive stimulation (COGSTIM) and age at kindergarten entry (AGEENTRY) are zero. A one unit increase in reading skills (READING) is associated with a 6.73 unit increase in cognitive stimulation (COGSTIM) and a 0.29 unit increase in age at kindergarten entry (AGEENTRY).

Y = -3.59 + 6.73X1 + 0.29X2

For COGSTIM, t-value = 35.10 with a significant p-value of 0.0000000000000002. For AGEENTRY, t-value = 14.81 with a significant p-value of 0.0000000000000002.

Standardized

```
data_1 <- data_1 %>%
  mutate(zREADING = scale(READING))
data_1 <- data_1 %>%
  mutate(zCOGSTIM = scale(COGSTIM))
data_1 <- data_1 %>%
  mutate(zAGEENTRY = scale(AGEENTRY))
head(data_1)
```

```
##    CHILDID AGEENTRY BLACK COGSTIM READING   zREADING  zCOGSTIM  zAGEENTRY
## 1 0001002C       74     0    2.68   45.17  2.0944250 1.3404760  1.9138975
## 2 0001007C       60     0    2.34   25.67 -0.5555363 0.5566649 -1.4048881
## 3 0001010C       67     0    2.91   27.34 -0.3285909 1.8707012  0.2545047
## 4 0002003C       71     0    2.25   42.51  1.7329431 0.3491855  1.2027291
## 5 0002004C       63     0    2.61   36.15  0.8686480 1.1791032 -0.6937198
## 6 0002006C       59     0    2.31   22.59 -0.9740943 0.4875051 -1.6419442
```

```
model_1_z <- lm(zREADING ~ zCOGSTIM + zAGEENTRY, data= data_1)
summary(model_1_z)
```

```
##
## Call:
## lm(formula = zREADING ~ zCOGSTIM + zAGEENTRY, data = data_1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.2584 -0.6442 -0.1235  0.5380  3.5571
##
## Coefficients:
##                          Estimate            Std. Error t value
## (Intercept) 0.0000000000000004961 0.0112947735692862285    0.00
## zCOGSTIM    0.3965127992600874229 0.0112961817766425218   35.10
## zAGEENTRY   0.1672713481070050301 0.0112961817766424576   14.81
##                        Pr(>|t|)
## (Intercept)                   1
## zCOGSTIM    <0.0000000000000002 ***
## zAGEENTRY   <0.0000000000000002 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9035 on 6396 degrees of freedom
## Multiple R-squared:  0.1839, Adjusted R-squared:  0.1837
## F-statistic: 720.7 on 2 and 6396 DF,  p-value: < 0.00000000000000022
```

```
anova(model_1_z)
```

```
## Analysis of Variance Table
##
## Response: zREADING
##             Df Sum Sq Mean Sq F value                Pr(>F)
## zCOGSTIM     1  997.7  997.74 1222.22 < 0.00000000000000022 ***
## zAGEENTRY    1  179.0  179.00  219.27 < 0.00000000000000022 ***
## Residuals 6396 5221.3    0.82
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
confint(model_1_z, level=0.95)
```

```
##                   2.5 %     97.5 %
## (Intercept) -0.02214154 0.02214154
## zCOGSTIM     0.37436850 0.41865710
## zAGEENTRY    0.14512705 0.18941565
```

```
coefficients(model_1_z)
```

```
##               (Intercept)                  zCOGSTIM                 zAGEENTRY
## 0.00000000000000004961146 0.3965127992600874229367 0.1672713481070050300925
```

The intercept is 4.961146e-16 and is the predicted level of reading skills when cognitive stimulation (COGSTIM) and age at kindergarten entry (AGEENTRY) are zero. A one standard deviation unit increase in reading skills is associated with a 0.40 unit increase in cognitive stimulation and a 0.17 unit increase in age at kindergarten entry.

Y = 0.40X1 + 0.17X2

For COGSTIM, t-value = 35.10 with a significant p-value of 0.0000000000000002. For AGEENTRY, t-value = 14.81 with a significant p-value of 0.0000000000000002.

What proportion of the total variance in reading skills is explained by cognitive stimulation AND age at kindergarten entry? Adjusted R^2 so either is fine.

m1 <- lm(READING ~ COGSTIM + AGEENTRY, data= data_1) summary(m1) m1_r<-summary(m1) m1_r $r.squaredm1_r$ r.squared - summary(lm(READING~ READING, data=data_1))$r.squared # 0.18 (18%) represents the total variance in reading skills (READING) that is # explained by both cognitive stimulation (COGSTIM) and age at kindergarten # entry (AGEENTRY).

Is the proportion in c significantly different from zero?

resid(model_1) model_1.res <- resid(model_1) plot(data_1$READING, model_1.res)

```
m1 <- lm(READING ~ COGSTIM + AGEENTRY, data= data_1)
summary(m1)
```

```
##
## Call:
## lm(formula = READING ~ COGSTIM + AGEENTRY, data = data_1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
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##
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```

```
m1_r<-summary(m1)
m1_r$r.squared
```

```
## [1] 0.1839225
```

```
m1_r$r.squared - summary(lm(READING~ READING, data=data_1))$r.squared
```

```
## Warning in model.matrix.default(mt, mf, contrasts): the response appeared on
## the right-hand side and was dropped
```

```
## Warning in model.matrix.default(mt, mf, contrasts): problem with term 1 in
## model.matrix: no columns are assigned
```

```
## [1] 0.1839225
```

R^2 = variance explained by the model/total variance

Yes 0.18 is significantly different from zero because our F statistic is above zero [F(2,6396)=720.7] and our p-value is significant (p=0.00000000000000022).